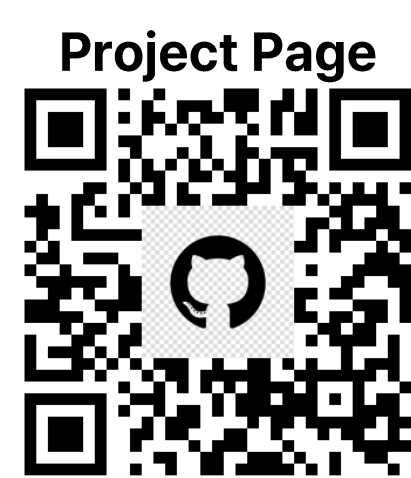


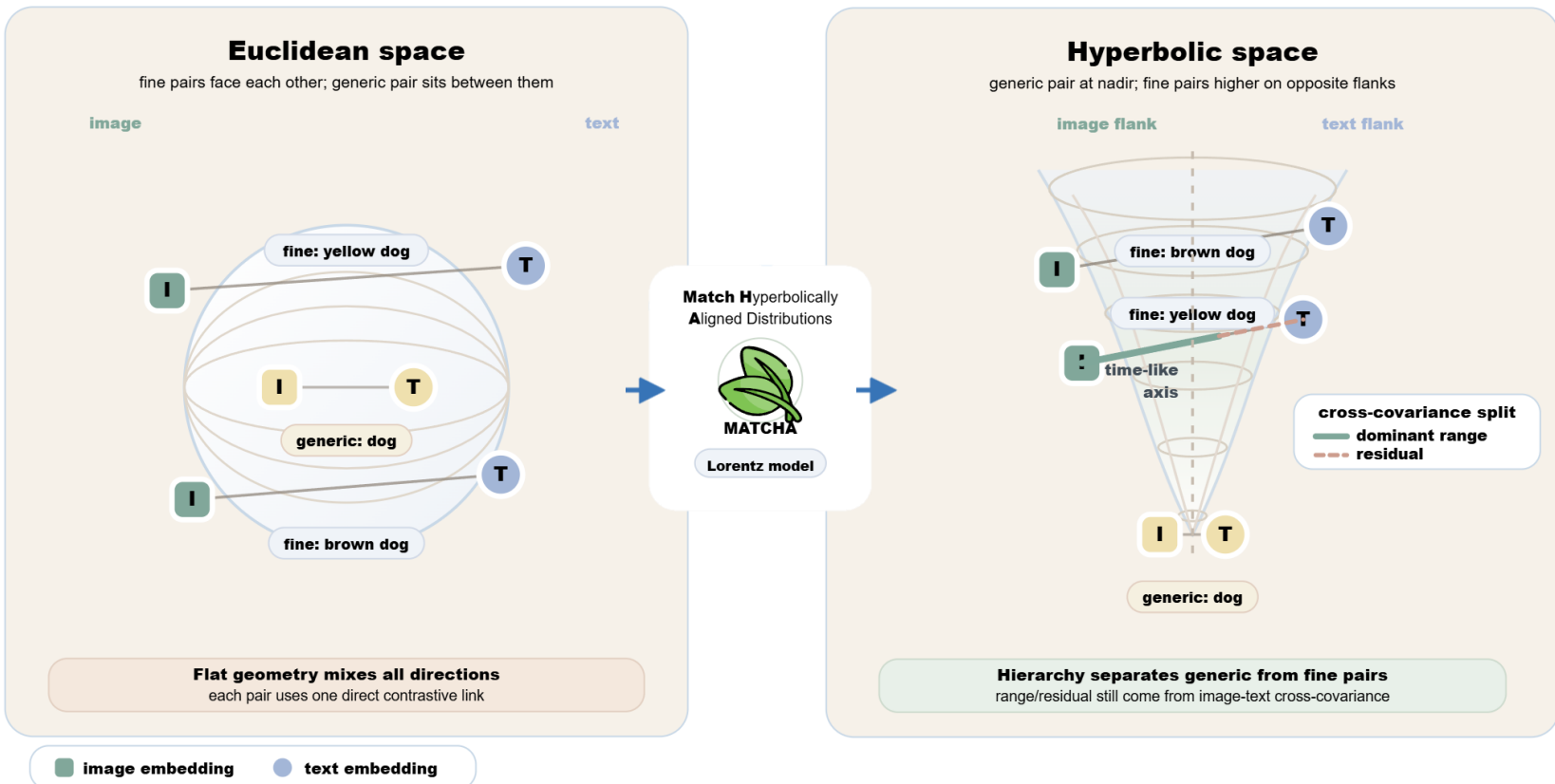


Introduction

- Demand for multimodal datasets for efficient training of VLMs is growing!
- Existing vision-language dataset distillations preserve *cross-correlation* structure, yet **miss explicit pairwise structure** and does not consider the hierarchy in text
→ **RAHA**: hierarchical structure + relative similarity order

Motivation

Transition to Hyperbolic Space:



What must data compression preserve?

- A distilled image-text set is useful only if a model trained on it learns the same bidirectional ranking behavior as from real data. This makes **cross-modal relevance** the core information capacity to preserve, beyond visual appearance alone.

How should text hierarchy be retained?

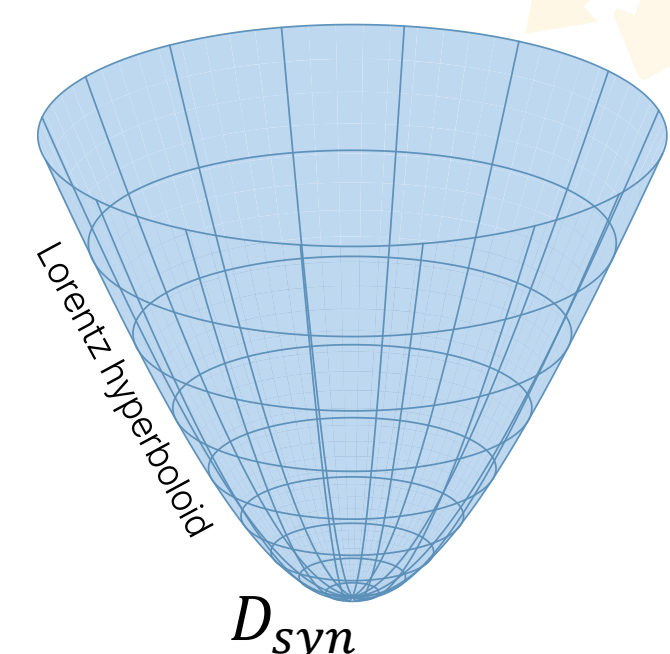
- Since captions move from generic categories to fine attributes and relations, a geometry that keeps **generic text broad**, and **fine-grained distinct** on a hyperboloid is beneficial

Where should alignment capacity be spent?

- Previous study has shown that **useful image-text relations** are concentrated in **low-rank** structure. In RAHA, we separates range and residual subspaces, matching dominant shared semantics while regulating weaker residual variation.

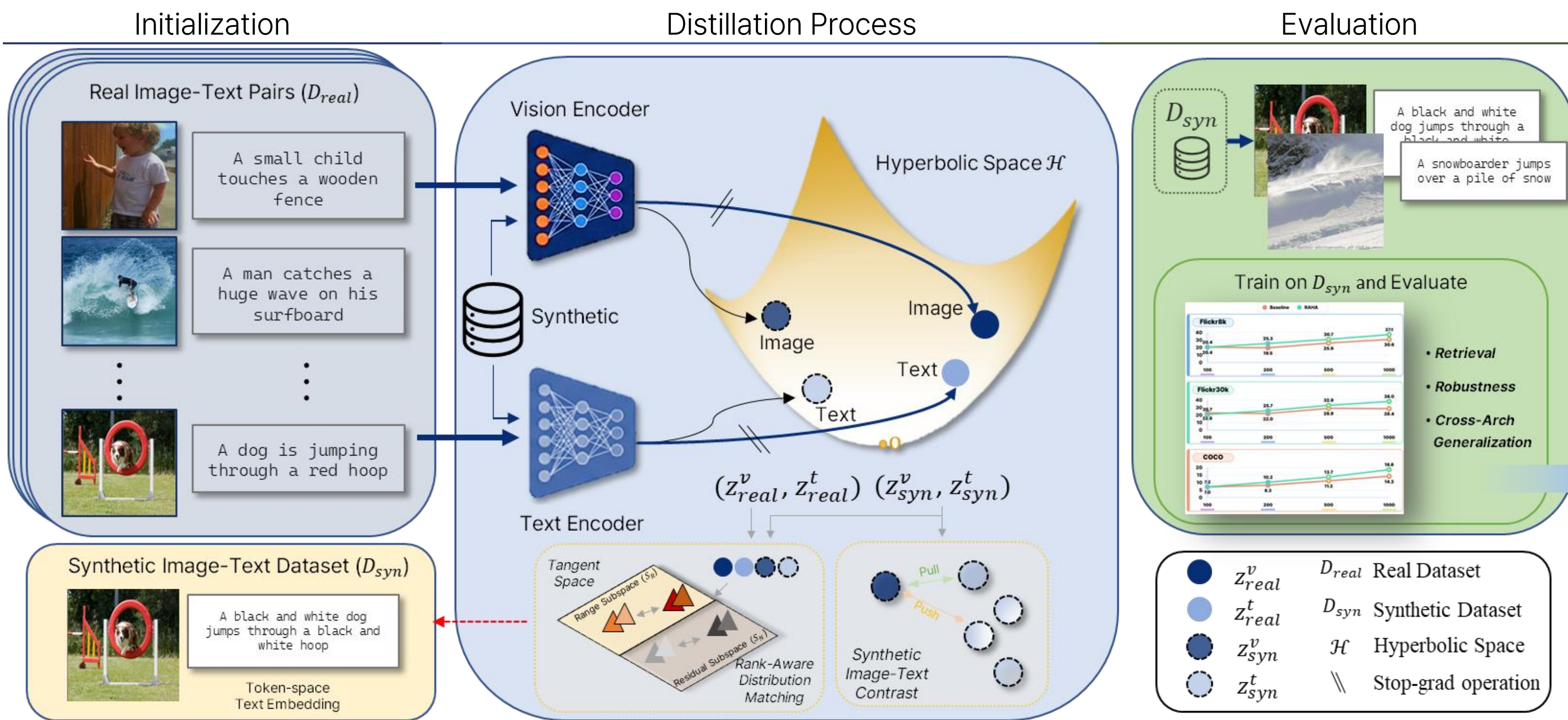
Advantages of hyperbolic embeddings

- Can map more **generic** data **closer** to the origin, more **specific** data **farther** away from the origin,
- Conceptually, $\left[\text{distance from the origin} \right] \propto \left[\text{uncertainty of inputs from inherent noisy image-text pairs} \right]$



Proposed Method: RAHA

Match Hyperbolically Aligned Distributions!



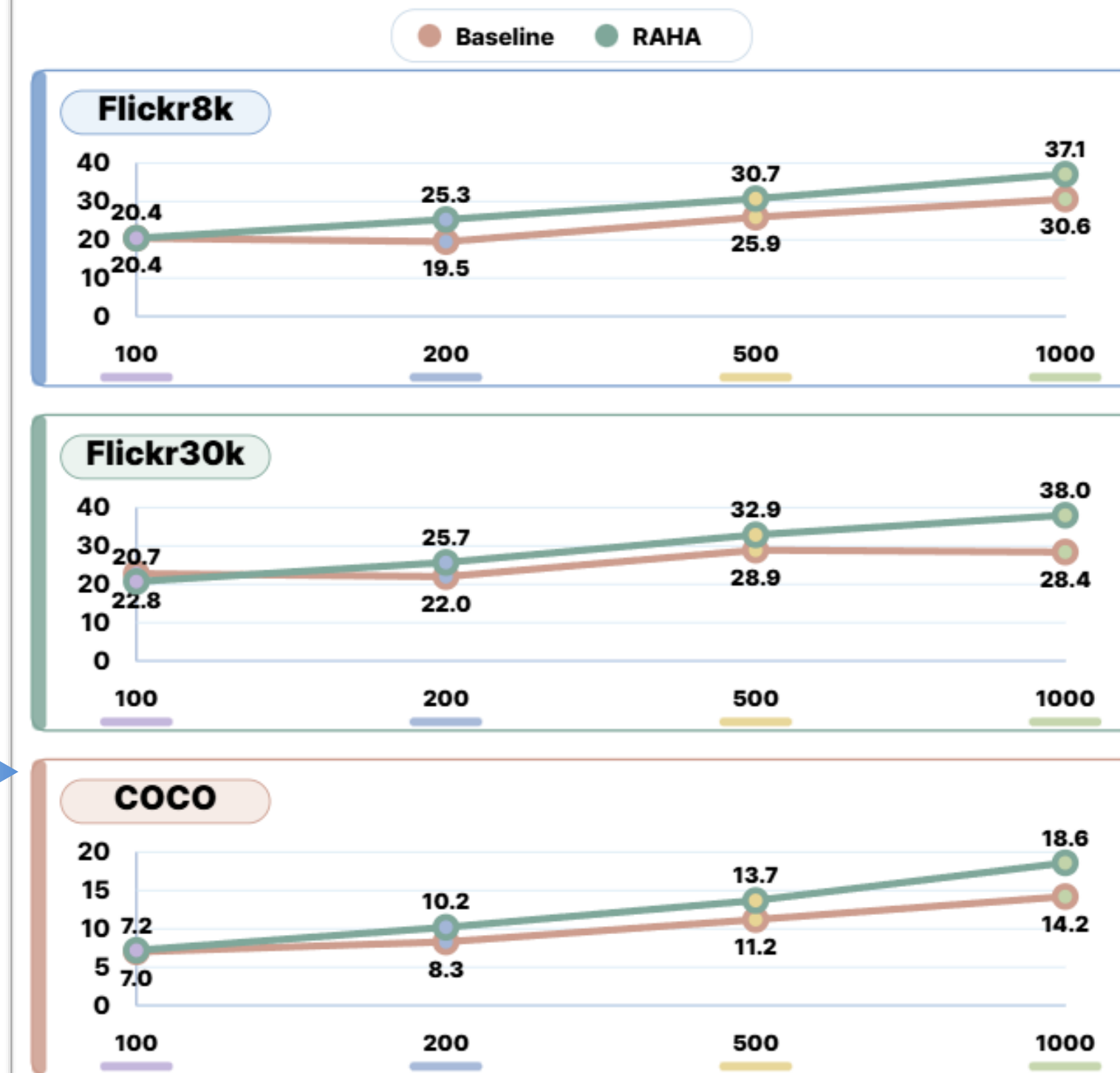
Key Contributions

- Rank-aware hyperbolic formulation for vision-language dataset distillation
- Range-residual subspace-decomposition for cross-modal relevance distillation with optimal transport
- Comprehensive evaluation on retrieval, cross-arch. transfer, robustness, prompted classification

Qualitative Comparison

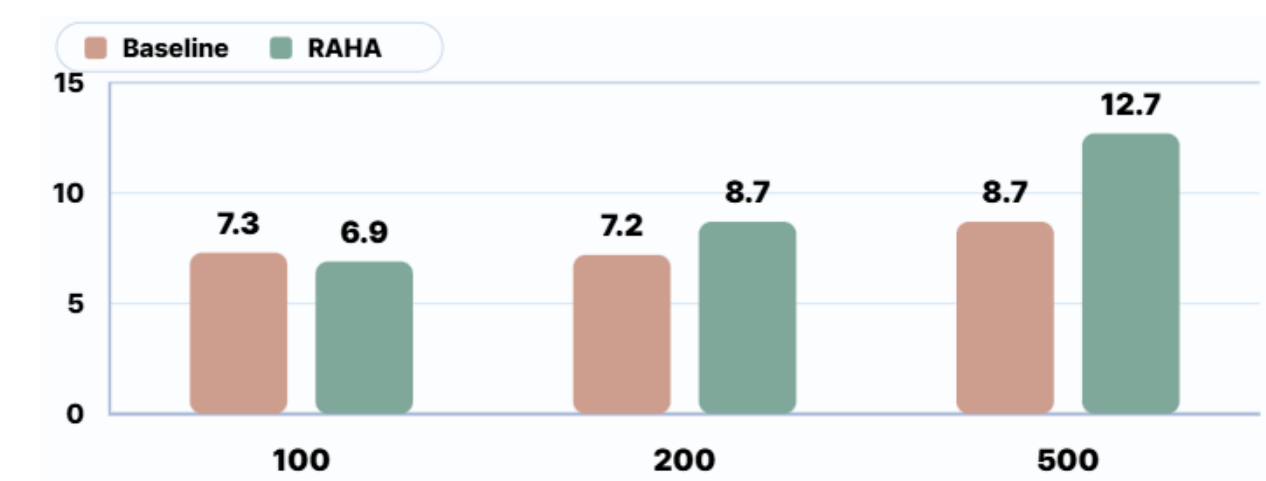


Experiments



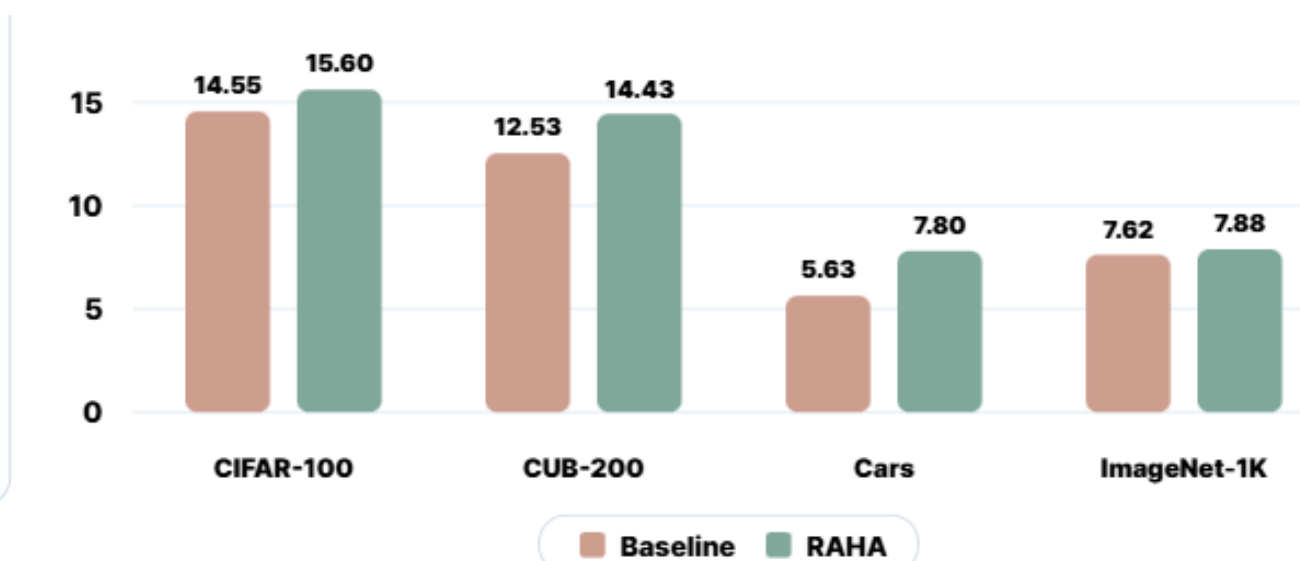
DATASET	PAIRS	BASLINE MEAN	RAHA MEAN	Δ
Flickr8k	100	20.4	20.4	+0.0
	200	19.5	25.3	+5.8
	500	25.9	30.7	+4.8
	1000	30.6	37.1	+6.5
Flickr30k	100	22.8	20.7	-2.1
	200	22.0	25.7	+3.7
	500	28.9	32.9	+4.0
	1000	28.4	38.0	+9.6
COCO	100	7.0	7.2	+0.2
	200	8.3	10.2	+1.9
	500	11.2	13.7	+2.5
	1000	14.2	18.6	+4.4

Cross-Architecture Transfer

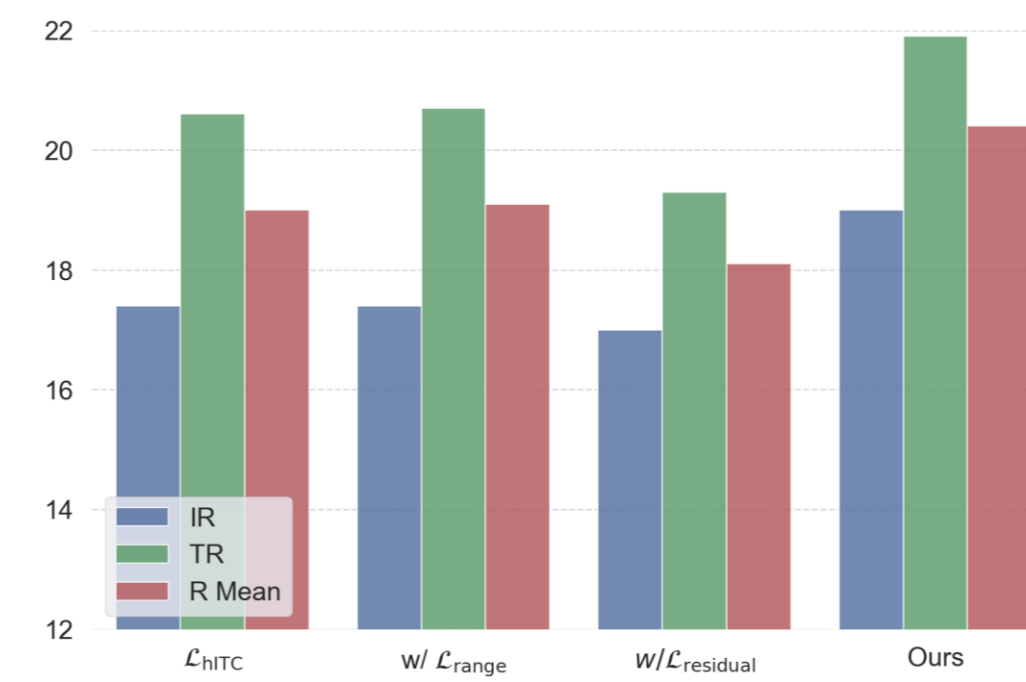


*Averaged over {NF-ResNet, NF-RegNet, ViT-B}, {BERT, DistilBERT} w/o source model

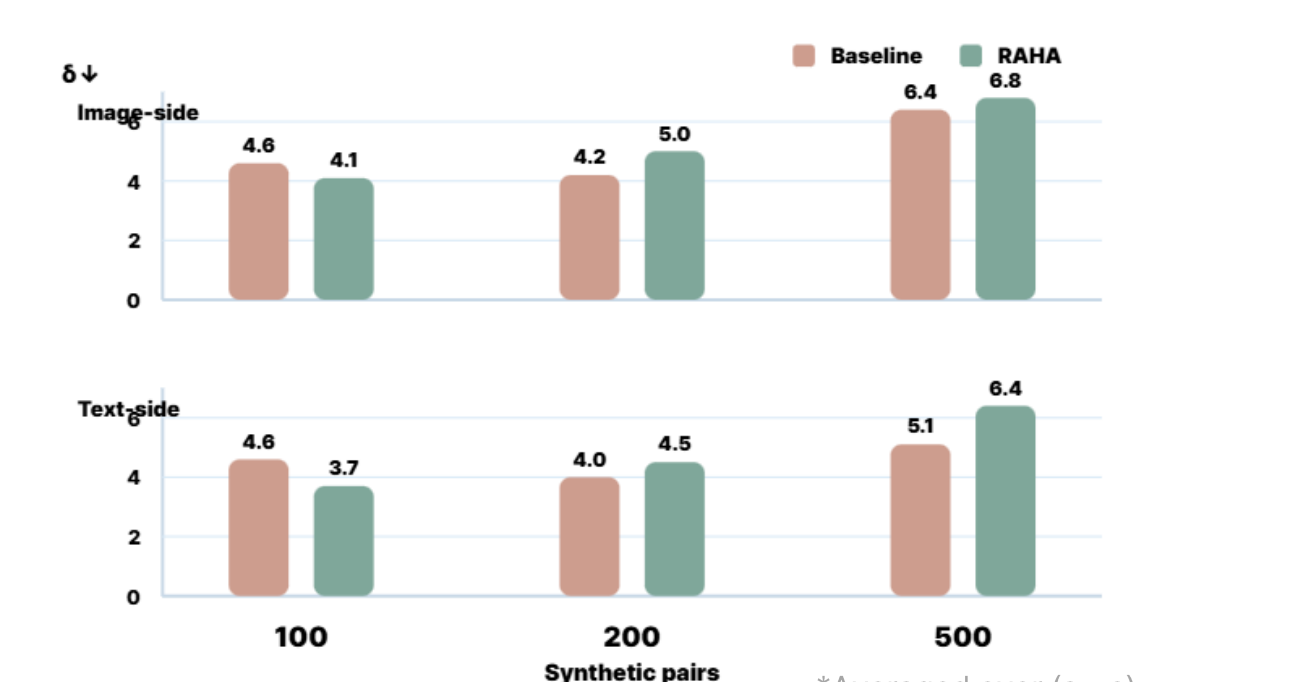
Prompted Image Classification



Component Analysis



Robustness to Noise



(a) JPEG (75%)/Bit Quantization (4-bit)
 (b) Additive White Gaussian Noise (AWGN) ($\sigma = 0.01$)
 (c) 10-PGD (step size=2)

*Averaged over (a-c)